

AIRRRPrep: Workflows for Preparing Bulk and Single-Cell AIRR-Seq Data for Annotation

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Abstract

Summary: AIRRRPrep is a web application for reproducible preprocessing of bulk and single-cell AIRR-seq data, providing predefined and customizable pRESTO workflows with input-aware compatibility checks and stepwise provenance, as well as harmonization of seven single-cell input specifications with linked retention of available source annotations. Validation on public datasets showed record-level agreement with equivalent pRESTO workflows and preservation of the tested sequence-to-cell metadata.

Availability and Implementation: AIRRRPrep is freely available without registration at <https://preprocessing.computimm.com>. The Python backend, React frontend, container definitions and validation resources are available under the GNU Affero General Public License v3.0 or later at <https://github.com/ComputImm/AIRRRPrep>. Release v1.0.0 is archived at <https://doi.org/10.5281/zenodo.22857985>.

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Supplementary information: Supplementary data are available at *Bioinformatics* online.

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Introduction

Adaptive immune receptor repertoire sequencing (AIRR-seq) data enter analysis as bulk reads or heterogeneous single-cell inputs. Here, preprocessing refers to preparing receptor sequences and associated metadata for subsequent V(D)J annotation or reannotation and repertoire analysis. For bulk inputs, this involves configurable read-processing operations. For single-cell inputs, it involves receptor reconstruction when required and format and metadata harmonization (hereafter, harmonization) of sequences, cell identifiers and available metadata across supported input formats. This preparation is necessary because inputs arrive at different processing stages, carry different information and cannot simply be treated as interchangeable.

Bulk reads may require quality filtering, primer masking, unique molecular identifier (UMI) extraction, paired-read assembly, consensus generation and duplicate collapsing before V(D)J assignment. These operations are order- and state-dependent, requiring particular file types, read-stream states, annotations and companion files. Single-cell data may instead comprise raw reads, reconstructed sequences or platform-specific contig and annotation files; harmonization must preserve available sequence-cell links and source annotations without inventing missing identifiers. Errors in either path can propagate downstream.

The increasing adoption of AIRR-seq across experimental systems has created a growing need for flexible and reproducible preprocessing

strategies. Unlike conventional sequence-processing pipelines in which individual steps can often be considered independently, AIRR-seq preprocessing operations are linked through evolving data states. The output of one operation determines the valid inputs of subsequent operations, including available annotations, file formats, read relationships and companion files. Consequently, workflow errors may remain hidden until late stages of analysis or may produce outputs that are technically generated but computationally inconsistent. This challenge is amplified in single-cell AIRR-seq, where receptor sequences must remain linked to their originating cells while integrating outputs from multiple sequencing and analysis platforms.

Existing software addresses these stages at different levels. pRESTO provides command-line preprocessing of raw AIRR-seq reads (Vander Heiden et al., 2014). TRUST4 reconstructs receptor sequences from bulk and single-cell RNA-seq (Song et al., 2021), whereas nf-core/airrflow provides a configuration-driven, end-to-end workflow for both bulk and single-cell AIRR-seq (Gabermet et al., 2024). Web platforms have broader scopes: VDJServer extends from read preprocessing and V(D)J gene assignment to repertoire analysis and data sharing (Christley et al., 2018), while ARGalaxy analyzes and visualizes annotated repertoires (IJspeert et al., 2017).

These tools solve complementary parts of AIRR-seq analysis but leave different responsibilities to the user. Direct use of pRESTO requires users to maintain compatible file and header-annotation states

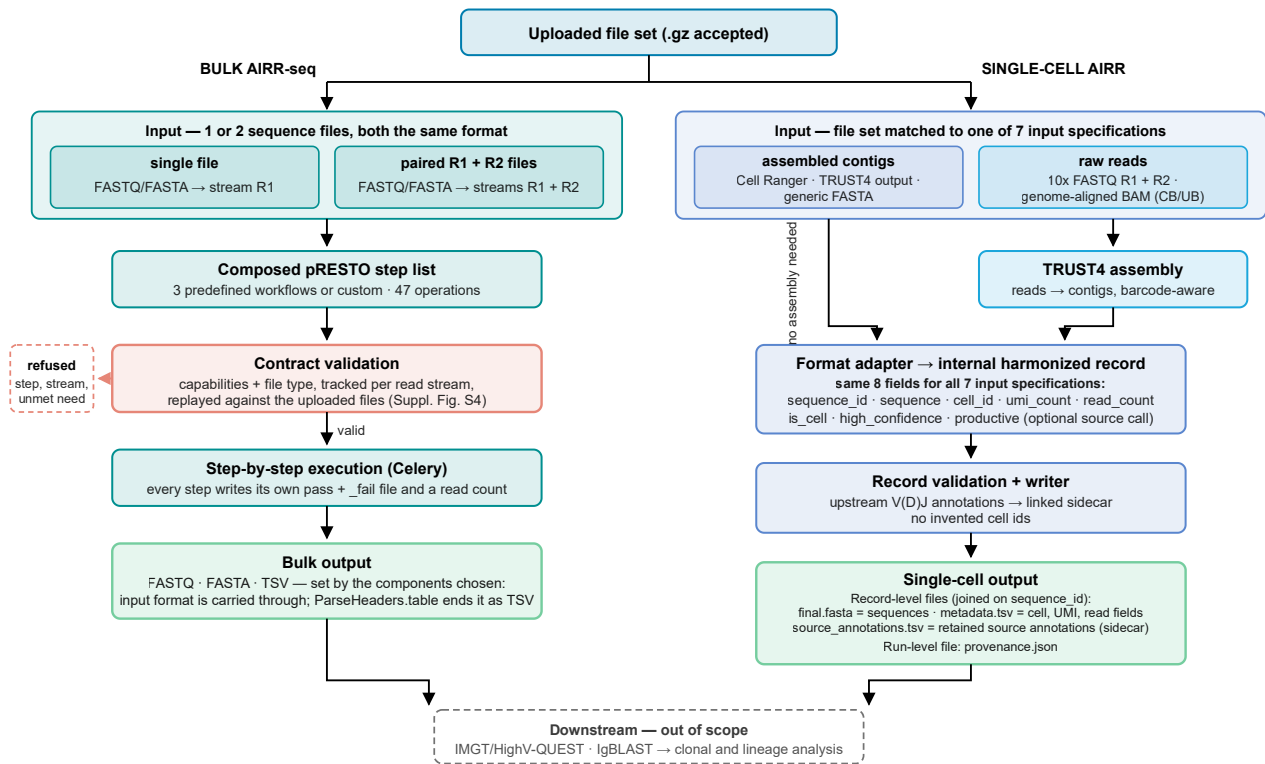


Figure 1 Overview of AIRRPrep. Bulk inputs are processed using predefined or custom pRESTO workflows whose computational compatibility is checked before execution. Single-cell inputs are harmonized directly when reconstructed sequences are provided or are first assembled with TRUST4 when assembly is required. Available source annotations are retained in a linked sidecar keyed by the same `sequence_id` as the harmonized output.

and to order interdependent operations correctly. End-to-end workflows improve portability and reproducibility, but follow predefined or configuration-driven analysis paths rather than supporting interactive composition of individual pRESTO operations. Documented web platforms emphasize broader end-to-end workflows or downstream repertoire analysis. Across the systems and documentation examined, we did not identify a platform that combined input-aware validation of custom pRESTO workflows before execution with harmonization of heterogeneous single-cell inputs while retaining available source annotations in linked outputs (Supplementary Table S11).

AIRRPrep was designed to provide this combination within a single web application. For bulk data, it offers three predefined protocols and a builder covering 47 pRESTO operations. State-aware contracts track file type, required annotations, companion files and R1/R2 state during construction and before execution, identifying computationally incompatible workflows before they run. For single-cell data, AIRRPrep accepts reconstructed inputs or invokes TRUST4 when assembly is required, then maps seven supported input specifications to a common sequence-and-metadata representation. It does not invent missing cell identifiers and retains available source annotations in a linked sidecar keyed by the same `sequence_id`. Workflow graphs, per-step record counts, provenance and exportable definitions support inspection and rerunning. These checks establish computational coherence, not the biological appropriateness of parameter choices.

Implementation

AIRRPrep provides distinct processing routes for bulk and single-cell AIRR-seq data, as summarized in Figure 1.

Bulk workflow construction and pre-execution validation

For bulk processing, AIRRPrep represents input as either one read stream from a FASTA or FASTQ file or separate R1 and R2 streams from matched paired-end files of the same format. Three predefined workflows reproduce documented pRESTO examples: UMI-barcoded MiSeq 2×250 (Stern et al., 2014), non-UMI MiSeq 2×250 (Greiff et al., 2014) and UMI-barcoded 5' RACE MiSeq 325+275 (Vander Heiden et al., 2017); full parameters are listed in Supplementary Table S9. A custom builder lets users compose workflows from 47 operations across 13 pRESTO tools (Supplementary Table S2). Each operation declares a contract specifying accepted file types, required and produced capabilities, companion files and read-stream behavior (Supplementary Figure S4). Whereas pRESTO validates individual commands at execution, AIRRPrep propagates file type, available annotations and R1/R2 state through the proposed workflow beforehand. It rejects the first incompatible step and reports the affected step, stream and unmet requirement, such as quality filtering on FASTA or consensus generation before barcode annotation. The same contracts update which operations are selectable during construction and explain why others are disabled, guiding users toward workflows that are computationally coherent but not necessarily biologically appropriate (Supplementary Note S1). The interface also displays a

count-annotated workflow graph, tracking R1 and R2 separately until assembly and the merged stream thereafter; the graph is exportable as a PNG (Supplementary Figures S2 and S3).

Single-cell format and metadata harmonization

Single-cell repertoire inputs contain receptor sequences together with cell-associated information, including cell barcodes, UMI counts and read support. AIRRPrep preserves available sequence-to-cell relationships while converting heterogeneous input formats into a unified representation for downstream repertoire analysis. Supported inputs include reconstructed outputs from Cell Ranger and TRUST4, as well as raw 10x FASTQ and genome-aligned BAM files (Supplementary Table S1). When reconstructed sequences are provided, AIRRPrep directly parses and harmonizes the existing output. For raw sequencing inputs, receptor assembly is first performed with TRUST4, followed by parsing and harmonization of the resulting sequences and associated cell-level information.

Format-specific adapters convert supported inputs into AIRRPrep's eight-field internal record, which is not an AIRR Rearrangement record. The run writes three record-level outputs: **final.fasta**, which carries the receptor sequences; **metadata.tsv**, which carries the available cell identifiers, UMI counts and read support; and **source_annotations.tsv**, the sidecar holding the retained source annotations. One run-level file, **provenance.json**, completes the output. Absent optional metadata are left as empty fields rather than inferred. The sequence-to-cell link is treated differently: cell identifiers are never generated from sequence identifiers alone, inputs lacking a valid cell mapping are refused, and TRUST4 contigs that the source assigns to no cell are excluded and counted (Supplementary Table S1). The optional **productive** field is copied from Cell Ranger only on request; it remains a source call and is not inferred by AIRRPrep. This approach maintains available links between receptor sequences and their originating cells while providing a consistent structure across input sources.

Upstream receptor annotations, including V(D)J gene calls, junctions and CDR3 annotations, are excluded from the harmonized representation to enable consistent downstream annotation using a defined reference and procedure, such as IMGT/HighV-QUEST (Alamyar et al., 2012). Available source annotations are retained separately in **source_annotations.tsv**, linked to output records through the same **sequence_id**. Processing information and checksums for the whole run are recorded in **provenance.json** (Supplementary Table S1).

Reproducibility and operation

Workflow definitions can be exported for reuse, while completed runs provide exportable records of workflow structure, outputs and execution metadata. Workflow graphs can be exported as images, allowing users to document the executed pipeline in manuscripts or reports. External companion files, such as primer FASTA inputs, are not embedded in workflow definitions; instead, required inputs are recorded with their file names, sizes and SHA-256 checksums. Exported records also include the software versions used during execution. Long-running jobs can be recovered using email-based tracking without requiring a persistent user account (Supplementary Note S2).

AIRRPrep uses a web-based client-server architecture with asynchronous job execution and state management through worker processes and Redis. The containerized environment includes TRUST4 (Song et al., 2021), BLAST+ (Camacho et al., 2009), samtools (Danecek et

al., 2021), MUSCLE (Edgar, 2004), VSEARCH (Rognes et al., 2016) and CD-HIT (Fu et al., 2012) and can be deployed on institutional infrastructure. The proprietary USEARCH is not included, and no predefined workflow depends on it. Software versions and execution dependencies are listed in Supplementary Table S10.

Validation

The three predefined bulk workflows were evaluated against equivalent command-line pRESTO implementations using the first 25,000 read pairs from SRR4026043 (Vander Heiden et al., 2017), ERR346600 (Greiff et al., 2014) and SRR1383456 (Stern et al., 2014). Across all three workflows, AIRRPrep and pRESTO produced identical records, sequences, quality strings and annotations at each compared step; only record ordering differed (Supplementary Tables S3–S5). Individual validation of the 47 supported operations showed equivalent output for 46 operations. The remaining **ConvertHeaders.py** **migec** case could not be directly compared because native pRESTO does not process the corresponding multi-record MIGEC input. The workflow validator accepted the predefined workflows and rejected invalid compositions involving missing annotations, incompatible file types, missing companion files and invalid read-stream states (Supplementary Note S1).

Single-cell harmonization was evaluated using Cell Ranger representations of the 10x Genomics **sc5p_v2_hs_PBMC_10k** T-cell V(D)J dataset. Harmonized outputs matched the reference records without sequence, cell barcode, UMI, read count or cell-status discrepancies (Supplementary Table S6). Generic FASTA inputs reproduced the same records when barcode mappings were provided, while inputs lacking a valid barcode mapping were rejected. TRUST4-based routes were evaluated using raw 10x reads, a contig-aligned BAM and a genome-aligned gene-expression BAM from the **sc5p_v2_hs_PBMC_1k** dataset. The raw-read and contig-aligned BAM routes recovered all 83 Cell Ranger-defined cells, whereas the genome-aligned gene-expression BAM route recovered 79 of 83; no route assigned artificial cell identifiers (Supplementary Table S7).

Execution time was evaluated on three replicates using identical pRESTO and Python environments on a Windows 11 workstation. AIRRPrep reduced execution time for the tested 25,000-read-pair workflows, primarily by avoiding repeated interpreter startup overhead between steps. Reference-guided assembly showed similar runtime between implementations because the underlying alignment operation dominated execution time. These measurements represent small-scale comparative execution tests on one workstation and are not intended as estimates of production-scale throughput.

Discussion

AIRRPrep simplifies access to established AIRR-seq preprocessing workflows while adding machine-checkable safeguards around workflow composition during construction and before execution. Stepwise record tracking makes runs auditable, and exported workflow definitions support reproducibility. Validation demonstrates computational coherence rather than biological appropriateness: a technically valid workflow can still contain unsuitable choices, such as an incorrect primer set or offset.

Read-pair assembly dominated the tested bulk workflows; TRUST4-based single-cell reconstruction was not included in the runtime

comparison. AIRRPrep is intentionally focused on preprocessing: it does not provide a separate downstream V(D)J reannotation stage, and its single-cell output is an internal representation rather than an AIRR Rearrangement record (Vander Heiden et al., 2018); therefore, AIRR compliance is not claimed. Future work will include cross-step quality-control summaries, support for additional single-cell platforms, resource-matched Linux benchmarks at production input sizes, and integration with downstream annotation tools.

Data availability

AIRRPrep is available as a free hosted service at <https://preprocessing.computimm.com>. The frontend and backend source code, container definitions, automated tests, validation scripts and their archived outputs (Supplementary Tables S2–S7), together with benchmark scripts and results (Supplementary Tables S8–S9), are available at <https://github.com/ComputImm/AIRRPrep> under the GNU Affero General Public License v3.0 or later. The version described in this manuscript is release v1.0.0 (commit 9148dba8bdd4c516b80db4eece07dd0a0b23a929), archived at <https://doi.org/10.5281/zenodo.22857985>.

Large validation and benchmark outputs that exceed repository limits, including per-operation validation outputs, single-cell validation outputs and benchmark run directories, are archived separately at <https://doi.org/10.5281/zenodo.22858375>. SHA-256 checksums are provided for archived files, and the repository links supplementary tables to the scripts and generated outputs used to produce them.

The bulk validation datasets consist of the first 25,000 read pairs from SRR4026043 (Vander Heiden et al., 2017), ERR346600 (Greiff et al., 2014) and SRR1383456 (Stern et al., 2014). Single-cell validation used Cell Ranger 5.0.0 outputs from the 10x Genomics sc5p_v2_hs_PBMC_10k and sc5p_v2_hs_PBMC_1k datasets. Input file checksums and validation outputs are provided in Supplementary Tables S6 and S7. USEARCH is proprietary and is not redistributed; the corresponding commands, software version and test results are provided for users with an appropriate licence.

Author contributions

Parsa Esmaeili: Conceptualization, Software, Validation, Writing – original draft. Fatemeh Chaker Hosseini Zavareh: Software, Methodology, Validation, Data curation, Writing – review & editing. Nika Abdollahi: Conceptualization, Methodology, Resources, Validation, Writing – review & editing. Changiz Eslahchi: Conceptualization, Supervision, Writing – review & editing.

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Conflict of interest

None declared.

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